

Discrete Mathematics
Assignment - 09 Sept, 2026

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* Can functions be applied to a "relation"? Justify your answer.

A: • Book: Discrete Mathematical Structures with Application to Computer Science - J.P. Tremblay, R. Manohar.

- Chapter 2-4: Functions.

- Section 2-4.2 - defines composition of functions as follows:

- Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two functions. The composite "relation" $g \circ f$ such that

$$g \circ f = \{ (x, z) \mid (x \in X) \wedge (z \in Z) \wedge (\exists y) (y \in Y \wedge y = f(x) \wedge z = g(y)) \}$$

is called the composition of "functions" of P^h ~~f~~ f and g .

- Here, the operation of composition of relations is extended to functions.

- Further in Section 2-4.4 on Binary and n -ary operations, an "operation" is defined as a mapping, such as a binary operation $f: X \times X \rightarrow X$.

- So if we take the example of the composite functions, where the operation of composition is performed, the function g operates on the function f , where f is ~~is~~ can be defined as a special relation.

- Hence, it can be stated that using an operation like composition on a set of functions means that 'a function is being applied to a relation'.